PIONEER JUNIOR COLLEGE
JC2 Preliminary Examination
In preparation for General Certificate of Education Advanced Level
Higher 2

BIOLOGY 9744/04

Paper 4 Practical **29 August 2017**

Junior College Year 2 **2 h 30 min**

CONFIDENTIAL INSTRUCTIONS

Question 1
Fresh G1, W, S1, S2 and Benedict's are needed for each candidate.
More of the solutions should be available if requested by candidates.
Solutions and reagents provided to the candidates should be supplied in a suitable beaker, or container, for removal of the solution using a syringe.

Summary of solutions and reagents

labelled contents hazard concentration / %	Contents	Hazard	Concentration / %	volume / cm3
G1	Glucose solution	none	4	15cm^3
S1	Glucose solution	none	1.5	10cm^3
S2	Glucose solution	none	0.1	10cm^3
W	Distilled water	none	-	50cm^3
Benedict's solution	Benedict's solution	[H] Harmful irritant	-	40cm^3

It is advisable to wear safety glasses/goggles when handling chemicals.

Preparation of solutions and reagents

(i) G1, at least 15 cm^3 of 4% glucose solution in a small beaker or container, labelled **G1**.
This is prepared by dissolving 4 g of glucose in a beaker or container with 80 cm3 of distilled water and making up to 100 cm3 with distilled water.
This is sufficient for 5 candidates.

(ii) S1, at least 10 cm3 of 1.5% glucose solution in a small beaker or container, labelled **S1**.
This is prepared by dissolving 1.5 g of glucose in a beaker or container with 80 cm^3 of distilled water and making up to 100 cm^3 with distilled water.
This is sufficient for 8-9 candidates.

(iii) S2, at least 10 cm3 of 0.1% glucose solution in a small beaker or container, labelled **S2**.

This is prepared by dissolving 0.1 g of glucose in a beaker or container with 80 cm^3 of distilled water and making up to 100 cm^3 with distilled water.
This is sufficient for 8-9 candidates.

(iv) W, at least 30 cm^3 of distilled water, in a beaker or container, labelled **W**.

(v) Benedict's solution, **[H]** at least 50 cm^3 of Benedict's solution, in a small beaker or container (so that a 10 cm^3 syringe can be used), labelled **Benedict's solution**.
These solutions can be made up the day before the examination and stored in a refrigerator. However, these must be at room temperature for the examination.

Apparatus for each candidate

Apparatus Quantity	**Quantity**
10 cm3 syringe or one with the means to wash it out	2
Container with tap water, labelled **washing**	1
Paper towels	4
Glass vials to hold 20 cm^3 volume	5
Test-tubes – large suitable for heating	8
Test-tube rack or containers to hold at least eight test-tubes	1
Water-bath equipment	1
Big beaker for water bath	1
Bunsen burner, tripod, gauze, bench mat, at least a 400 cm3 beaker with water suitable for a water bath (at approximately 70°C), matches and a thermometer –10 °C to 110 °C	1
Stop clock, stop watch or sight of a clock with a second hand	1
Glass marker pen	1
Safety goggles/glasses	1
Glass rod	1

During the examination, the Supervisor should, **out of the sight of the candidates**, carry out **Question 1** using the same solutions and reagents as the candidates. These results should be written in the Supervisor's report (**not** on a spare Question paper) which should be enclosed with the candidates' scripts. Please ensure that if the scripts are in several packets that a copy of the Supervisor's report is enclosed with each packet of scripts. The Invigilator should **not** carry out **Question 1**.

Question 2
Apparatus for each candidate
(i) Slide K**2** (supplied by Cambridge)
(ii) Microscope with:
• Low-power objective lens, e.g. × 4 AND X 10
• High-power objective lens, e.g. × 40
• Eyepiece graticule (supplied by Cambridge) fitted within the eyepiece and visible in focus at the same time as the specimen.
Please check that they are labelled **K2** and that all the slides are intact.

Question 3
There are no requirements for this question.